

Three Digital-Embryo Studies on Real Data: Turing Pattern Selection, Gene-Network Attractors, and Morphogen Positional Information

MEGA-27 Program, Project 14

September 2026

Abstract

We build and validate three computational embryology studies against real, public, accession-level data. Study A simulates Turing reaction–diffusion morphogen patterning (Gierer–Meinhardt kinetics) and measures a systematic, statistically robust *beyond-onset wavenumber upshift*: simulated patterns select wavenumbers 15–60% above the linear fastest-growing mode, with the largest deviations at intermediate distance from onset; we show honestly that this shift is *not* predictable from the linear dispersion relation by a convolutional surrogate ($n = 171$ simulations). Study B analyses the published *Drosophila* segment-polarity Boolean network (Cell Collective id-021) by exact attractor enumeration under all eight external signalling conditions, reproduces the published bistability, quantifies the wild-type basin as 0.77% of the synchronous state space, and measures the sample complexity of learning the basin boundary (graph neural network vs. multilayer perceptron). Study C analyses 676 real Bicoid-GFP morphogen gradients and 2,180 gap-gene immunofluorescence embryo records (Liu, Morrison & Gregor, PNAS 2013): we (i) reproduce the gradient length scale under the authors’ own fitting window, (ii) quantify a dosage-coupled length-scale effect that is concentrated in the anterior embryo (Spearman $\rho = 0.36$, $p = 8 \times 10^{-16}$ in anterior windows; absent in the authors’ posterior window), (iii) beat the published threshold model of cephalic-furrow positioning with a one-dimensional convolutional decoder (RMSE 2.38% egg length vs. 4.94% egg length, held-out fly lines; paired bootstrap CI for the gain [1.99, 3.11]% EL), and (iv) show that positional decodability from Hb+Eve expression improves with maternal *bcd* dosage (6.21% vs. 9.91% EL median error; bootstrap CI [3.00, 4.40]% EL). Negative results (a retracted small-sample surrogate win, single-exponential sufficiency, degenerate basin classification) are reported alongside positive ones. All code, tests, data manifests and result files are public at the project repository.

Contents

- 1 Introduction**
 - 2 Mathematical preliminaries and derivations**
 - 3 Study A: Turing pattern selection**
 - 4 Study B: Gene-regulatory-network attractors**
 - 5 Study C: Morphogen gradients and positional information**
 - 6 Tools, datasets and reproducibility**
 - 7 Honest negatives and limitations**
 - 8 Conclusion**
- References**